One variable (Simple) Linear Regression

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LSM

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad i = 1, \dots, n$$

$$\mathbb{E}\varepsilon_i = 0$$

Standard Assumption: ε_i independent, $\operatorname{Var}\varepsilon_i = \sigma^2$

β_1 slope, β_0 intercept

Estimation

OLS $\quad L_2(u_0, u_1) = \sum_i (Y_i - u_0 - u_1 X_i)^2$

LAD $\quad L_1(u_0, u_1) = \sum_i |Y_i - u_0 - u_1 X_i|$

If we further assume that $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$ then we can also use MLE

log likelihood $\ell(\beta_0, \beta_1, \sigma^2)$

$$= -\frac{1}{2\sigma^2} \sum (Y_i - \beta_0 - \beta_1 X_i)^2 - \frac{n}{2} \log \sigma^2 - c(n)$$

log likelihood under Double Exponential

$$\longrightarrow -\frac{1}{k\sigma} \sum |Y_i - \beta_0 - \beta_1 X_i| - n \log \sigma - c(n,k)$$

OLS estimator

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{(u_0, u_1)}{\operatorname{argmin}} \underbrace{\sum (Y_i - u_0 - u_1 X_i)^2}_{L(u_0, u_1)}$$

Normal Equation $\quad \frac{\partial L}{\partial u_0} = 0, \quad \frac{\partial L}{\partial u_1} = 0$

$$\begin{cases} \sum (Y_i - u_0 - u_1 X_i) = 0 \\ \sum (Y_i - u_0 - u_1 X_i) X_i = 0 \end{cases}$$

$$\underbrace{\begin{pmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{pmatrix}}_{\text{non-singular if } \det(\,) \neq 0} \begin{pmatrix} u_0 \\ u_1 \end{pmatrix} = \begin{pmatrix} \sum Y_i \\ \sum X_i Y_i \end{pmatrix}$$

$$\parallel \quad \mathbb{X}^T\mathbb{X}$$

i.e. $n\sum x_i^2 \neq \left(\sum x_i\right)^2$

true if not all x_i's same.

i.e. if $X = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix} \neq c\mathbb{1}_n \qquad \mathbb{X} = \begin{pmatrix} 1 & X_1 \\ \vdots & \vdots \\ 1 & X_n \end{pmatrix}$

Note:

1. $\hat{\beta}_0 + \hat{\beta}_1 \bar{X} = \bar{Y}$ —— ①

2. $\hat{\beta}_0 \sum X_i + \hat{\beta}_1 \sum X_i^2 = \sum X_i Y_i$ —— ②

$\hat{\beta}_0 \bar{X} + \hat{\beta}_1 \overline{X^2} = \overline{XY}$

$\Rightarrow \hat{\beta}_1 \frac{1}{n}\sum X_i^2 - \bar{X}^2 = \frac{1}{n}\sum X_i Y_i - \bar{X}\bar{Y}$

$\Rightarrow \hat{\beta}_1 = \dfrac{\frac{1}{n}\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\frac{1}{n}\sum (X_i - \bar{X})^2} = \dfrac{S_{XY}}{S_X^2}$

$\sum_i (Y_i - u_0 - u_1 X_i)^2$

$= \sum_i \Big((Y_i - \bar{Y}) - \underbrace{(u_0 - \bar{Y} + u_1 \bar{X})}_{= v_0} - \underbrace{u_1}_{v_1} (X_i - \bar{X}) \Big)^2$

$= \sum_i \left(\tilde{Y}_i - v_0 - v_1 \tilde{X}_i \right)^2$

$\begin{pmatrix} n & \cancel{\sum \tilde{X}_i}\, 0 \\ \cancel{\sum \tilde{X}_i}\, 0 & \sum_i \tilde{X}_i^2 \end{pmatrix} \begin{pmatrix} v_0 \\ v_1 \end{pmatrix} = \begin{pmatrix} \sum \tilde{Y}_i \\ \sum \tilde{X}_i \tilde{Y}_i \end{pmatrix} \Rightarrow \begin{array}{l} v_0 = 0 \\ v_1 = \dfrac{\sum \tilde{X}_i \tilde{Y}_i}{\sum \tilde{X}_i^2} \end{array}$

Effectively we perform an orthogonalization operator

$$\mathbb{X} = (\mathbb{1}_n : X)$$

$$\tilde{\mathbb{X}} = (\mathbb{1}_n : X - \bar{x}\mathbb{1}_n)$$

$$\underline{\tilde{x}} = x - \bar{x}\underline{\mathbb{1}}_n$$

$$= \left(I_n - \frac{1}{n}\mathbb{1}_n\mathbb{1}_n^T\right)\underline{x} = P_{\mathbb{1}_n}^{\perp}$$

orthogonal projection matrix orthogonal to the space spanned by $\mathbb{1}_n$

$\beta = \begin{pmatrix}\beta_0\\ \beta_1\end{pmatrix}$ M1 $\underline{Y} = \mathbb{X}\underline{\beta} + \varepsilon = \beta_0\underline{1} + \beta_1\underline{X} + \varepsilon$

M2 $\underline{Y} = \tilde{\mathbb{X}}\gamma + \varepsilon$

$$= (1 : \tilde{x})\begin{pmatrix}\gamma_0\\ \gamma_1\end{pmatrix} + \varepsilon \qquad \gamma_0 = \beta_0 + \bar{x}\beta_1$$

$$\gamma_1 = \beta_1$$

$$= \gamma_0\underline{1} + \gamma_1(\underline{x} - \bar{x}\underline{1}) + \varepsilon$$

$$\tilde{\mathbb{X}}^T\tilde{\mathbb{X}} = \text{diag}\underline{o}\text{nal} = \begin{pmatrix} 1^T1 & \cancel{1^T\tilde{x}}^{\,0} \\ \cancel{1^T\tilde{x}}^{\,0} & \tilde{x}^T\tilde{x}\end{pmatrix}$$

$$\tilde{x} = P_{\mathbb{1}}^{\perp}x \Rightarrow \mathbb{1}^T\tilde{x} = \mathbb{1}^T P_{\mathbb{1}}^{\perp}\underline{x} = 0 \cdot \underline{x} = 0$$

So diagonal

M3 $\underline{\tilde{Y}} = \tilde{\mathbb{X}}\delta + \varepsilon \qquad \underline{\tilde{Y}} = \underline{Y} - \underline{\bar{Y}}\mathbb{1}_n$

$$= \mathbb{X}\beta - \bar{Y}\mathbb{1}_n + \varepsilon \quad \text{M4}$$

$$Y = \mathbb{X}\beta \qquad \tilde{Y} = \left(I - \frac{1}{n}\mathbb{1}_n\mathbb{1}_n^T\right)\underset{\sim}{Y} = P_1^{\perp}Y$$

$$P_1^{\perp} Y = P_1^{\perp} \mathbb{X}\beta + P_1^{\perp}\varepsilon$$ (M5) (Not the original model, β_0 is missing)

LS estimate for β_1 from (M4) is

$$\tilde{Y} = \beta_1 \tilde{x} + \tilde{\varepsilon} \qquad \tilde{\beta}_1 = \frac{\sum \tilde{Y}_i \tilde{X}_i}{\sum \tilde{X}_i^2} = \frac{\sum (Y_i - \bar{Y})(X_i - \bar{X})}{\sum (X_i - \bar{X})^2} = \hat{\beta}_1$$

<u>Properties of the estimate:</u>

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$$

$$\hat{\beta}_1 = \frac{1}{n}\sum_{i=1}^{n} \frac{(X_i - \bar{X})}{S_X^2} Y_i, \quad S_X^2 = \sum (X_i - \bar{X})^2$$

$(\hat{\beta}_0, \hat{\beta}_1)$ is a linear function of $\underset{\sim}{Y}$

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = (\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T Y = (\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T(\mathbb{X}\beta + \varepsilon)$$
$$= \beta + (\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T\varepsilon$$

$$\mathrm{Var}(\underset{-}{\hat{\beta}}) = \mathrm{Var}(\underbrace{(\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T}_{B}\underset{-}{\varepsilon}) = B\, \underbrace{\mathrm{Var}(\underset{-}{\varepsilon})}_{=\sigma^2 I_n \text{ under the std. assumption}} B^T$$

$$= \sigma^2 B B^T$$

$$= \sigma^2 (\mathbb{X}^T\mathbb{X})^{-1}$$

If further we assume that $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

then $\underset{-}{\hat{\beta}} = \underset{-}{\beta} + \underset{2\times n}{B}\,\underset{n\times 1}{\underset{-}{\varepsilon}} \sim N(\underset{-}{\beta}, \sigma^2(\mathbb{X}^T\mathbb{X})^{-1})$

Sampling dist^{n} of $\hat{\beta}$ under Gaussian assumption

Confidence interval for β_j $j=0,1$ $\Gamma = (X^T X)^{-1}$

Case 1: σ^2 known

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2 \Gamma_{j+1,j+1})$$

$100(1-\alpha)\%$ C.I. for β_j

$$P(l_j(\underline{X},\underline{Y}) \le \beta_j \le u_j(\underline{X},\underline{Y})) = 1-\alpha$$

$$\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{\Gamma_{j+1,j+1}}} \sim N(0,1) \leftarrow \text{free of the parameter}$$

$\Rightarrow$ We can take $l_j(X,Y) = \hat{\beta}_j + \sigma\sqrt{\Gamma_{j+1,j+1}}\,\Phi^{-1}(\alpha/2)$

$u_j(X,Y) = \hat{\beta}_j + \sigma\sqrt{\Gamma_{j+1,j+1}}\,\Phi^{-1}(1-\alpha/2)$

If σ^2 is unknown

$$\underbrace{\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{\Gamma_{j+1,j+1}}}}_{\text{want it to be parameter free}} = \frac{\overbrace{\hat{\beta}_j - \beta_j / \sigma\sqrt{\Gamma_{j+1,j+1}}}^{\text{free of parameter}}}{\sqrt{\hat{\sigma}^2/\sigma^2}}$$

$\rightarrow$ needs to be parameter free at least, but not sufficient. If independent, then true